

Shivram Nekkanti

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EDUCATION

Northeastern University (Boston, MA)

Expected: Dec 2026

Masters in Data Science

Coursework: Introduction to Database Management, Algorithms, Natural Language Processing, Machine Learning, Unsupervised Machine Learning and MLOps

EXPERIENCE

Data Scientist Co-op

Aug 2025 – Jan 2026

Institute of Experiential AI

Portland, ME

- Architected an LLM-assisted validation framework using Gemini 1.5 Pro on Vertex AI, reducing manual batch review time by 75% (8h to 2h) while maintaining rigorous quality standards through 120+ custom unit tests.
- Enabled analytics-ready access to 23M policy records by migrating to **PostgreSQL (AWS RDS) with AWS Glue ETL**, validating 99.97% integrity through automated checks and source-vs-target parity validation.
- Improved **identification of high-risk / uprating policies by 20%** by developing a predictive model (PyTorch DNN) with custom feature engineering on 850K historical policies, supporting more consistent operational decisioning.
- Scaled model delivery to production by **deploying Dockerized Flask inference APIs on SageMaker** serving 3,200 predictions/hour with p95 < 180ms, **using async processing + Redis caching** to meet performance and reliability constraints.
- **Partnered with cross-functional stakeholders** to align model outputs with business rules and define success criteria for iterative model improvements and release readiness.

Machine Learning Intern

Sept 2023 – Aug 2024

Indian Servers

Hyderabad, India

- **Improved chatbot response accuracy by 20%** (from 68% to 81.6% intent classification F1) across 1,000+ unique customer queries by fine-tuning DistilBERT on 35K labeled support tickets using Hugging Face Transformers.
- **Reduced prompt engineering iteration time by 55%** by implementing **LangChain PromptTemplates and SequentialChains** with 12 reusable components.
- **Decreased average customer response time by 15%** (from 190 sec to 155 sec) by **deploying Celery asynchronous task queues with Redis message broker**, processing 850 concurrent requests with 4 worker nodes.

PROJECTS

RAG-Powered Research Assistant | FastAPI, Qdrant, LangChain, Ollama, Git, 

Dec 2024 - Jan 2025

- Built a citation-backed **RAG system using LangChain** pipelines and retrieval chains, integrating Hugging Face Sentence-Transformer embeddings to achieve 98% semantic retrieval accuracy.
- Implemented LangChain document loaders, text splitters, and retrievers to optimize vector indexing and improve relevance on long-context academic PDFs.
- Reduced retrieval latency by 60% by combining LangChain's asynchronous execution framework with a **Dockerized FastAPI + Qdrant deployment**.

TheraBot: Mental Health Chatbot | BERTweet, RoBERTa, DistilGPT2, Flask, Streamlit  **Jan 2025 – Mar 2025**

- Deployed a real-time chatbot with emotion and sarcasm detection for empathetic mental health support.
- Fine-tuned BERTweet with attention pooling for emotion classification (72% macro F1 on GoEmotions).
- Created a RoBERTa-based sarcasm detector with 74% accuracy on SARC.
- Developed a distilGPT2 response generator conditioned on emotion, sarcasm, and context.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (Postgres), R

Big Data Analytics: PySpark, Snowflake

Statistical Analysis: Hypothesis testing, Statistical modeling, A/B testing.

Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, SciPy, Seaborn, NLTK, spaCy, openCV

CI/CD & DevOps: Git, GitHub, Docker

Cloud Platforms: AWS (Lambda, S3, EC2, SageMaker, Bedrock, RDS), GCP

Machine Learning: Regression, Naive Bayes, Gradient Boosting, Random Forests, SVMs, Deep Learning